FIITJEE

JEE (Advanced)-2018

ALL INDIA TEST SERIES

FULL TEST – III

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C.**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section–A (01 – 08, 21 – 28, 41 – 48)** contains 24 multiple choice questions which have one or more correct answer. Each question carries **+4 marks** for correct answer and **–2 marks** for wrong answer.

   Section–A (09 – 12, 29 – 32, 49 – 52) contains 12 paragraphs with each having 2 questions with one correct answer. Each question carries **+4 marks** for correct answer and **–2 marks** for wrong answer.

2. **Section–C (13 – 20, 33 – 40, 53 – 60)** contains 24 Numerical based questions with answers as numerical value from 0 to 9 and each question carries +4 marks for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

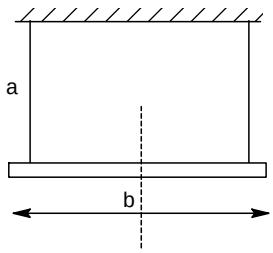
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **8** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A thin rod of length 'b' is suspended horizontally using ideal strings tied to both ends of the rod. The length of the strings is 'a'. The rod is given an initial angular speed ω about its central axis. Let δy be the upwards displacement of rod's centre in a small time interval δt and δF be the total increment in the tension forces just after the rod was given the angular speed.



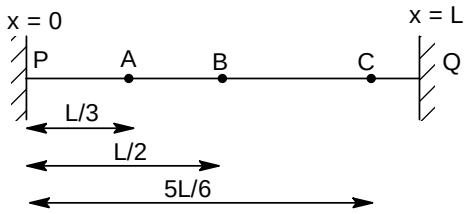
(A) $\delta y = \frac{b^2}{8a}(\omega\delta t)^2$

(B) $\delta y = \frac{b^2}{2a}(\omega\delta t)$

(C) $\delta F = \frac{mb^2}{4a}\omega^2$

(D) $\delta F = \frac{mb^2\omega^2}{8a}$

2. Standing waves are established on a string of length L such that A is a node and B is an immediate anti node. Oscillation amplitude for point B is a_0. Let a' be the oscillation amplitude for point C. Pick the suitable option(s) for correct value of a' and the possible equation(s) for standing waves on the string.



(A) $a' = \frac{a_0}{2}$

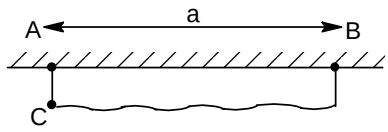
(B) $y = a_0 \sin\left(\frac{6\pi}{L}x\right)\cos(\omega t)$

(C) $a' = a_0$

(D) $y = a_0 \sin\left(\frac{3\pi x}{L}\right)\sin(\omega t)$

Space for Rough work

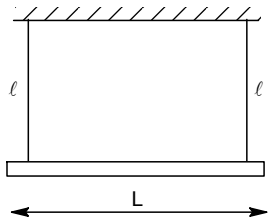
3. A string of length '2a' has been tied at A and B such that AB = 'a'. The string is passing through a bead C and initially the bead is very close to A. Now the bead is allowed to fall. Let h be the height by which the bead falls when the string becomes taut and v be the speed of the bead just after the string becomes taut. Pick correct option(s):—



(A) $v = \sqrt{3ag}\sin(26.5^\circ)$ (B) $v = \sqrt{1.5ag}\sin(26.5^\circ)$

(C) $h = \frac{3a}{5}$ (D) $h = \frac{3a}{4}$

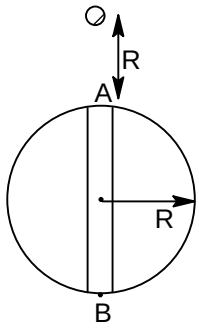
4. A rod of lengths L is supported by two ideal strings of length ℓ such that the system hangs in a vertical plane.
Case 1: Rod is kept horizontal, displaced slightly perpendicular to the plane and allowed by oscillate.
Case 2: Rod is given a small twist about central axis and then allowed to oscillate.
Let T_1, T_2 be the periods of oscillations in the two respective cases.



(A) $T_1 = 2\pi\sqrt{\frac{\ell}{g}}$ (B) $T_1 = 2\pi\sqrt{\frac{\ell}{2g}}$

(C) $T_2 = 2\pi\sqrt{\frac{L}{6g}}$ (D) $T_2 = 2\pi\sqrt{\frac{\ell}{3g}}$

5. A small particle is dropped from a height R in front of a narrow tunnel dug inside the earth (along a diameter). Let M be the mass of earth, R be radius of earth. Let v_0, T be speed of particle when it reaches A and time taken by particle to go from A to B respectively. Assuming mass of particle to be negligible as compare to mass of earth, pick the correct option(s)

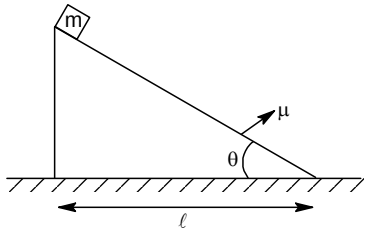


(A) $V_0 = \sqrt{\frac{GM}{2R}}$ (B) $V_0 = \sqrt{\frac{GM}{R}}$

(C) $R = \frac{\pi}{2}\sqrt{\frac{R^3}{GM}}$ (D) $T = \pi\sqrt{\frac{R^3}{GM}}$

Space for Rough work

6. A block of mass m slides down an incline plane. The incline plane has fixed base length 'ℓ' and coefficient of friction on the incline plane is 'μ'. The plane is fixed and block slides from top to bottom. Let θ_0 be the inclination angle for minimum sliding time and v_0 be the block's speed when it reaches the bottom in that case. Pick the correct option(s):—



(A) $v_0 = \sqrt{2g\ell(\tan\theta_0 - \mu)}$ (B) $v_0 = \sqrt{2g\ell\tan\theta_0}$

(C) $\tan(2\theta_0) = \frac{-1}{\mu}$ (D) $\tan(\theta_0) = \frac{1}{\mu}$

7. In Bohr model of hydrogen atom, let R, v and E represent the radius of robit, speed of the electron and total energy of electron respectively. Which of following quantities are directly proportional to quantum number n?

(A) vR (B) RE

(C) $\frac{v}{E}$ (D) $\frac{R}{E}$

8. A neutron having kinetic energy E_0 collides with singly ionised He atom at rest and move along initial direction. Which of the following statement(s) is/are true for the above mentioned collision.

(A) Collision will be perfectly inelastic if E_0 = 17 eV

(B) Collision will be perfectly inelastic is E_0 = 8.16 eV

(C) If E_0 = then ploton of wavelength λ = 500 nm is observed just after the collision if E_0 = 8.16 eV.

(D) Perfectly elastic collision is not possible.

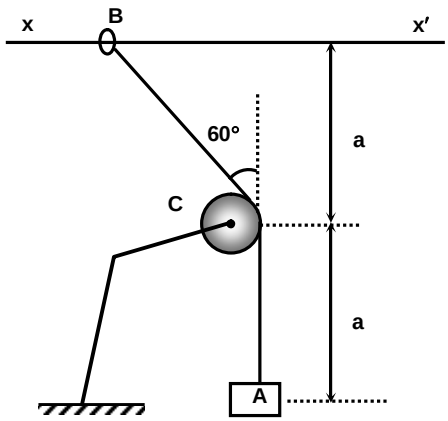
Space for Rough work

Paragraph type

This section contains **2 paragraphs**, each describing theory, experiments, data etc. Four questions relate to the two paragraphs with **two questions on each paragraph**. Each question has **ONLY ONE** correct answer among the four given options **(A), (B), (C) and (D).**

Paragraph for Question Nos. 09 and 10

A bead B of mass 'm' can travel without friction on a smooth horizontal wire xx′. The bead is connected to a block of identical mass by an ideal string passing over an ideal pulley. The system, as shown, is in vertical plane.



9. System is initially released from rest in the position shown. Initial acceleration of block A is:

(A) zero (B) $\frac{g}{2}$

(C) $\frac{3g}{7}$ (D) $\frac{4g}{7}$

10. The system is allowed to fall. Find the speed of block A at the instant when string connected to the bead B makes an angle 37° with vertical.

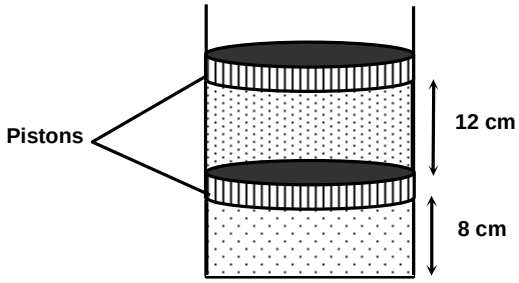
(A) $\sqrt{ag}$ (B) $\sqrt{\frac{5ag}{17}}$

(C) $\sqrt{\frac{27ag}{68}}$ (D) $\sqrt{\frac{ag}{2}}$

Space for Rough work

Paragraph for Question Nos. 11 and 12

Figure shows a long cylindrical container with ideal gas in two chambers. Lower chamber is filled with one mole of a mono atomic gas, while upper chamber has one mole of a diatomic gas. The gases initially are at temperature 300 K, the container walls as well as pistons are conducting. Both the pistons are identical with mass 'M' and area 'A' such that $\frac{Mg}{A} = P_0$ (atmospheric pressure). Assuming the ideal gas constant to be R, answer the following questions:



11. The upper piston is pulled up slowly by 16 cm and held there. The displacement of the lower piston till it reaches new equilibrium state is:
(A) 2 cm (B) 4 cm
(C) 8 cm (D) 12 cm

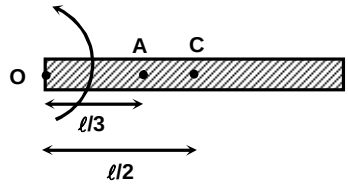
12. Total work done by the ideal gases in this process is
(A) 300 Rℓn(3) (B) $300R\ell n\left(\frac{9}{5}\right)$
(C) zero (D) 300 Rℓn(2)

Space for Rough work

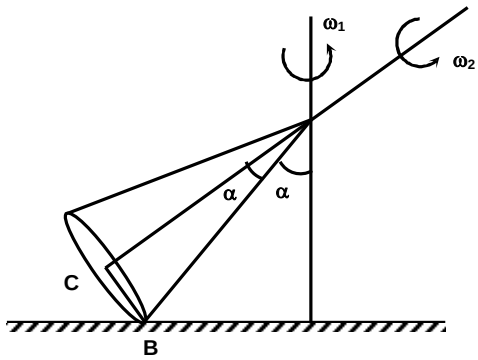
SECTION – C
(Single digit integer type)

This section contains **8** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

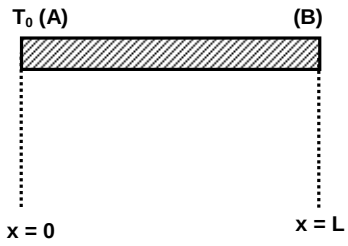
13. A uniform rod of mass m, length ℓ rotates about its end point O in horizontal plane. If the rod is rotating with a constant angular speed on a frictionless surface and the ratio of restoring force developed in the rod at points A and C is $\frac{F_A}{F_C} = 1 + \frac{n}{27}$, where 'n' is an integer value, find n.



14. A uniform solid cone (height h = 4m, $\alpha = 30^o$) is inclined against a vertical axis as shown in the figure. The cone rotates about its own axis as well as rotates about the vertical axis with angular speeds marked in the diagram. If the cone does not slip at point B; find $\frac{\omega_1}{\omega_2}$.

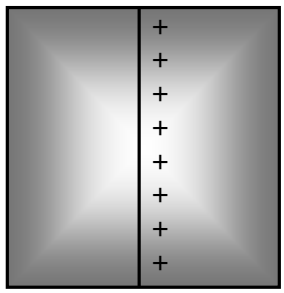


15. Figure shows a metal rod of uniform cross section area A, with variable thermal conductivity given by $k(x) = k_0 \sec\left(\frac{\pi}{6L}x\right)$. If the end A is maintained at temperature T_0, the rod carries a thermal current I_0 (from B to A) in steady state and $\frac{I_0 L}{k_0 A T_0} = \frac{\pi}{3}$; find the temperature of the end B of the rod. Let's say this temperature is kT_0, find integer value k.

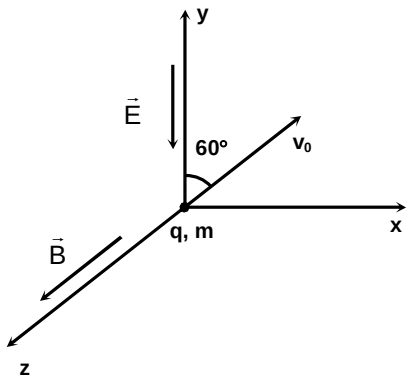


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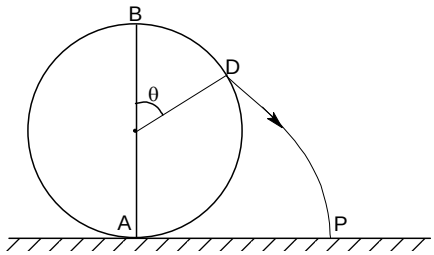
16. A very large non-conducting plane carrying charge Q uniformly distributed over it is placed in a medium with resistively ρ and permittivity ε. The plane leaks charge through the surrounding medium and the charge becomes $\frac{1}{n}$ times the initial value after a time $t = 6\rho\,\varepsilon\,\ell n(2)$ elapses. Find the integer value n. (Neglect the end effects).



17. A positively charged particle is given an initial velocity v_0 in XY plane as shown in diagram. Initially the particle was at origin and the region contains uniform, mutually perpendicular electric and magnetic field. Let $\vec{E} = -E\hat{j}$ and $\vec{B} = B\hat{k}$. The minimum magnitude of the magnetic field so that the charge particle crosses the origin again during its subsequent motion is $B_{min} = \frac{n\pi E}{v_0}$, find the integer value n.

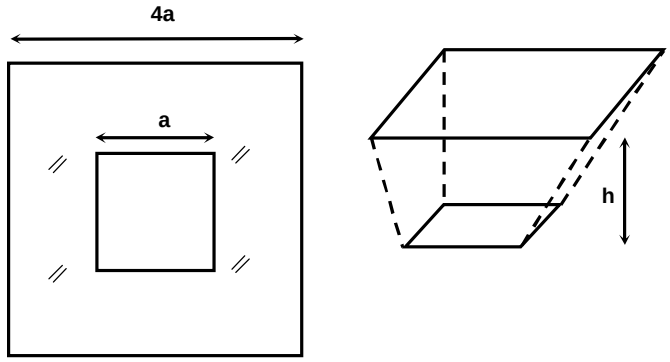


18. A particle slides down the surface of a smooth fixed sphere of radius R starting from rest at the highest point B. Particle leaves the sphere at some point and then strikes the horizontal plane passing through the lowest point A of the sphere, at point P. The distance AP is given by $AP = \frac{n}{27}[\sqrt{n} + 4\sqrt{2}]R$ where 'n' is an integer, find n. (Take $g = 10\ m/s^2$).



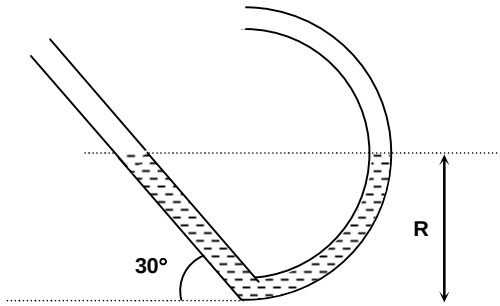
Space for Rough work

19. Figure shows a soap film formed between two square figures made of a uniform wire. The bigger square is held while keeping it in a horizontal plane and the smaller square is slowly allowed to drop vertically. It reaches an equilibrium state after dropping a height h. Let surface tension of soap = T. Mass per unit length of wire = λ. Acceleration due to gravity = g.



Given that $h = \dfrac{n\lambda ga}{2\sqrt{4T^2 - \lambda^2 g^2}}$; find the integer value n.

20. A tube with thin but uniform cross section has two arms, one straight, other shaped as a semicircle of radius r. Initially both arms carry an ideal fluid upto a height R. Now the equilibrium is disturbed by pushing the fluid in the left arm by a small amount. Fluid is then released and allowed oscillate. Neglect any friction or viscous forces. If the time period of oscillations is found to be



$T = \pi\sqrt{\dfrac{nR}{3g}(\pi + n)}$; find the integer value n.

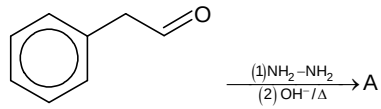
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Chemistry

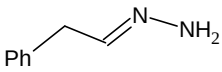
PART – II

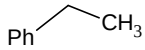
SECTION – A
(One or More than one correct type)

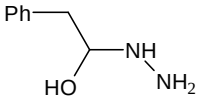
This section contains **8** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

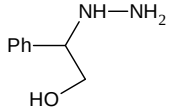
21. 

A is

(A) 

(B) 

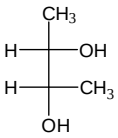
(C) 

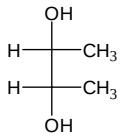
(D) 

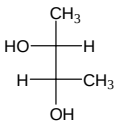
22. 100 ml of 0.05 M $CuSO_4$(aq) solution was electrolysed using inert electrodes by passing current till pH of the resulting solution was 2. The solution after electrolysis was neutralized and then treated with excess of KI and formed I_2 titrated with 0.04 M $Na_2S_2O_3$. Calculate the required vol. in ml of $Na_2S_2O_3$.

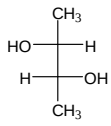
(A) 112.5 ml (B) 100 ml
(C) 125 ml (D) none of these

23. Which of the following structure(s) is/are meso-2,3 butanediol?

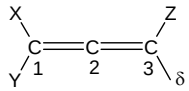
(A) 

(B) 

(C) 

(D) 

Space for Rough work

24. 

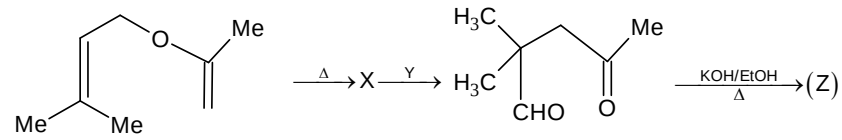
The π electron cloud of $C_1 - C_2$ is present in the plane of paper than which of the following is incorrect?

(A) Z is perpendicular to the plane of paper.

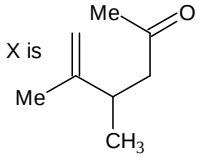
(B) X is present in the plane of paper.

(C) σ bond of $C_2 - C_3$ is perpendicular to the plane of paper.

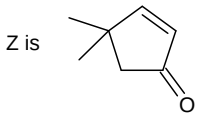
(D) π electron cloud of $C_2 - C_3$ bond and X is present in same plane.

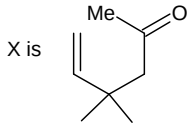
25. 

Choose the correct option (s)

(A) 

(B) Y is $O_3/(CH_3)_2S$

(C) 

(D) 

26. How many of the following reaction(s) have sum of co-efficient of reactants after balancing either 3 or 4.

(A) $FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$

(B) $Br_2 + \bar{O}H \longrightarrow BrO^- + Br^- + H_2O$

(C) $Cl_2 + SeO_3^{2-} + H_2O \longrightarrow SeO_4^{2-} + Cl^- + H^+$

(D) $Cl_2 + IO_3^- + OH^- \longrightarrow Cl^- + IO_4^- + H_2O$

27. $(COOH)_2 \xrightarrow{heat} X(gas) + Y(gas) + Z(gas)$

Y and Z both are polar and neutral, X is non polar and acidic. Z gas is condensed and formed liquid having pH = 7. The hybridization state of X, Y and Z are respectively.

(A) sp, sp^2, sp^3

(B) sp^2, sp^3, sp^3d

(C) sp, sp, sp^3

(D) sp^2, sp^2, sp^3

28. Which of the following give no residue on heating?

(A) NH_4NO_3

(B) NH_4NO_2

(C) $(NH_4)_2Cr_2O_7$

(D) NH_4Cl

Space for Rough work

Paragraph type

This section contains **2 paragraphs**, each describing theory, experiments, data etc. Four questions relate to the two paragraphs with **two questions on each paragraph**. Each question has **ONLY ONE** correct answer among the four given options **(A), (B), (C) and (D).**

Paragraph for Question Nos. 29 to 30

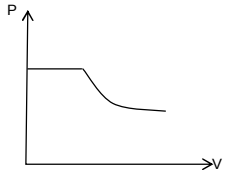
When triatomic gas X_3 reacts with an excess of potassium iodide solution buffered with a borate buffer (pH = 9.2), diatomic product Y_2 is liberated which can be titrated against a standard solution of sodium thiosulphate. This is a quantitative method for the estimation of X_3 gas in the mixture of X_3 and X_2 of some atom X.

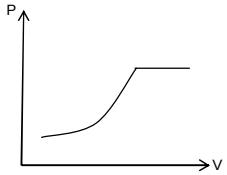
29. X_2 and Y_2 are respectively?
(A) Cl_2 and I_2 (B) O_2 and I_2
(C) N_2 and I_2 (D) O_2 and H_2

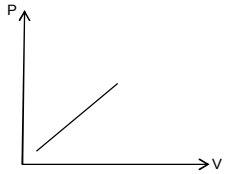
30. Select the incorrect statement?
(A) Y_2 produces blue colour with starch
(B) X_2 is thermodynamically more stable as compared to X_3
(C) Y_2 can produce brown colouration due to the presence of excess KI
(D) X_2 and X_3 both are colourless and odourless gases

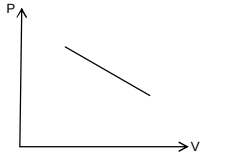
Paragraph for Question Nos. 31 to 32

2 moles of He gas $\left(r = \frac{5}{3}\right)$ are initially at a temperature of 27°C and occupy a volume of 20 L. The gas is first expanded at constant pressure until the volume is doubled, then undergoes reversible adiabatic change until the temperature returns to its initial value.

31. Which curve is correct for P vs V?
(A) 

(B) 

(C) 

(D) 

32. What are the initial pressure and final volume of gas?
(A) 2.46 atm, 110 L (B) 4.9 atm, 113.13 L
(C) 2.46 atm, 113.13 L (D) 4.9 atm, 110 L

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **8** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

33. Total number of moles of $EDTA^{4-}$ required to produce octahedral complex with Mg^{2+} is

34. Number of moles of NaOH required for complete neutralization of H^+ in solution which is formed by complete hydrolysis of 1 mole of PCl_5.

35. If 2s – 2p mixing is not operative the diamagnetic species among the following is/are
$H_2, H_2^+, Li_2, B_2, C_2, N_2, O_2^-, F_2$

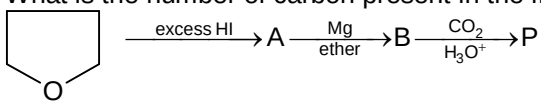
36. Find total number of d-electron(s) in metal ion complex $[M(NH_3)_4]^{2+}$. [EAN of metal = 35]

Space for Rough work

37. Equal volumes of 0.2 M KI solution and 0.1 M HgI_2 solution are mixed. Find the van't Hoff factor of the resulting solution.

38. In the monochlorination of 3-methylpentan, let x be the number of pairs of isomers which exist as enantiomers, y be the number of pairs of isomers which exist as diastereomers, z be the number of isomers which are achiral. Calculate the value of x + y + z.

39. What is the number of carbon present in the final product of given synthesis?



40. In 1 litre saturated solution of AgCl [$K_{sp} = 1.6 \times 10^{-10}$], 0.1 mole of CuCl [$K_{sp} = 1 \times 10^{-6}$] is added. The resultant concentration of Ag^+ in the solution is 1.6×10^{-x} M. The value of x is

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **8** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

41. Consider f as a twice differentiable function such that $f(x) + f''(x) = -x\, g(x) f'(x)\ \forall\ x \geq 0$ where, $g(x) \geq 0\ \forall\ x \geq 0$, then ($\forall\ x \geq 0$)
(A) $(f(x))^2 + (f'(x))^2$ is a non increasing function
(B) $(f(x))^2 < 3(f(0))^2 + (2f'(0))^2$
(C) $|f(x)| \leq \alpha$, α is a fixed real constant
(D) $\lim\limits_{x\to\infty} f(x)\cdot\sin\left(\frac{1}{x}\right)$ exist

42. Consider P, Q, R to be vertices with integral coordinates and $(|PR| + |RQ|)^2 < 8$. Area (ΔPQR) + 1, then
(A) $\angle R$ can be a right angle
(B) ΔPQR can be isosceles
(C) P, Q, R can lie on a square
(D) P, Q, R can lie on circle centred on midpoint of line segment PQ

43. Consider vectors $\vec{a}$, $\vec{b}$, $\vec{c}$; $\vec{p} = (\vec{b}\cdot\vec{c})\vec{a} - (\vec{c}\cdot\vec{a})\vec{b}$; $\vec{q} = (\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}$; $\vec{r} = (\vec{b}\cdot\vec{a})\vec{c} - (\vec{b}\cdot\vec{c})\vec{a}$, then
(A) $\vec{p}\cdot\vec{c} = 0$
(B) $\vec{p}, \vec{q}, \vec{r}$ can form a triangle
(C) $\Delta A(\vec{a})B(\vec{b})C(\vec{c})$ and $\Delta P(\vec{p})Q(\vec{q})R(\vec{r})$ are similar
(D) $\vec{p}, \vec{q}, \vec{r}$ are collinear

Space for rough work

44. Consider an equation $z^{1997} = 1$ (z is a complex number). If α, β are its two randomly chosen roots. 'p' denotes the probability that $\sqrt{2+\sqrt{3}} \le |\alpha+\beta|$, then

(A) $p = \frac{a}{1996}$, $a \in I$ (B) $\left[\frac{1}{p}\right]$ lies in [5, 15)

(C) $\left[\frac{1}{p} \times \frac{83}{50}\right]$ is 8 is divisible by 2 (D) $\left[\frac{1}{p}\right] + 2$ is 6

45. Let ABCD be a quadrilateral with $\angle CBD = 2\angle ADB$, $\angle ABD = 2\angle CDB$, AB = BC, then

(A) AD = CD (B) $\angle ADB = \angle CDB$

(C) $\angle CBD = \angle ABD$ (D) $\angle ADC$ is $\frac{2\pi}{3}$

46. If, f: $(0, \infty) \to (0, \infty)$ for which there is a positive real number 'a' such that it satisfies differential equation $f'\left(\frac{a}{x}\right) = \frac{x}{f(x)}$, then

(A) f(x) can be linear

(B) f(x) can be a functional of the type $m(x)^{1/n}$; $m \in R^+$, $n \in I^+$

(C) f′(x) can be positive

(D) f(x) can be twice differentiable

47. A variable plane passes through the point (1, 2, 3) and meets the coordinate axis is P, Q, R. Then, the locus of the point common to the planes through P, Q, R parallel to coordinate planes

(A) contains point (3, 6, 9) (B) passes through (0, 0, 0)

(C) is $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$ (D) contains line $\frac{x-1}{2} = \frac{y-2}{4} = \frac{z+6}{15}$

48. Consider a, b, c, d, e, f, g, h to be eight distinct alphabets then the number of ways in which they can be divided in 4 – parts is

(A) 1260 if exactly 2 – parts are equal

(B) 1701 in total

(C) 280 if its divided into two partitions of 3 and two of 1

(D) 210 if all parts are equal

Space for rough work

Paragraph type

This section contains **2 paragraphs**, each describing theory, experiments, data etc. Four questions relate to the two paragraphs with **two questions on each paragraph**. Each question has **ONLY ONE** correct answer among the four given options **(A), (B), (C) and (D).**

Paragraph for Question Nos. 49 to 50

Read the following write up carefully and answer the following questions:
Consider a function f: $R \to R$; $f(x^2 + y f(z)) = x f(x) + z f(y)\ \forall\ x, y, z \in R$

49. If $f(x) = 0, \forall\ x \in R$ is not considered a part of solution set, then
(A) $f(\alpha) < \alpha^4\ \forall\ \alpha \in (0, 1)$ (B) $f(\alpha) < \alpha^2\ \forall\ \alpha \in (0, 1)$
(C) $f(\alpha) = \alpha^3$ for some $\alpha \in R^+$ (D) $\lim\limits_{\alpha\to 0^+} \frac{f(\alpha)}{\alpha} < \lim\limits_{\alpha\to 0^+} \frac{\sin\alpha}{\alpha}$

50. If $g(x) = |f(x)| + f(|x|)$ then ($f(x) = 0\ \forall\ x \in R$ is not considered part of solution set)
(A) g(x) is not continuous $\forall\ x \in R$
(B) g(x) is differentiable except at two points $\forall\ x \in R$
(C) g(x) is differentiable at $x = \sin\alpha$ ($\alpha \neq n\pi$, $n \in I$)
(D) g(x) is non differentiable at integers

Paragraph for Question Nos. 51 to 52

Read the following write up carefully and answer the following questions:
Let $A_0, A_1, ... A_{n-1}$ be a n-sided polygon with vertices as $1, \omega, \omega^2, ..., \omega^{n-1}$. Let $B_0, B_1, ..., B_{n-1}$ be another polygon with vertices $1, 1 + \omega, 1 + \omega^2, ..., 1 + \omega^{n-1}$ $\left[\omega = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}\right]$

51. For n = 4, $\frac{\text{Ar.}(A_0, A_1, A_2, A_3)}{\text{Ar.}(B_0, B_1, ..., B_3)}$ is λ, then
(A) $[\lambda] > 3$ (B) $\frac{3\lambda}{2} \in I^+$
(C) $\frac{[\lambda]}{3} \in I^+$ (D) λ is irrational

52. $\frac{\text{Ar.}(A_0\ A_1\ A_2)}{\text{Ar.}(B_0\ B_1\ B_2)}$ is
(A) less than 1 (B) an irrational number
(C) lies in $(0, \pi)$ (D) less than $\frac{\pi}{2}$

Space for rough work

SECTION – C
(Single digit integer type)

This section contains **8** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

53. For continuous function f: [0, 1] $\to$ R; A = $\int_0^1 \left(x^2 \cdot f(x)\right)dx$; B = $\int_0^1 x \cdot \left(f(x)\right)^2 dx$ then, maximum value of A – B is λ where $\left[\frac{1}{4\lambda}\right]$ is _____

54. Let A be an n $\times$ n matrix such that A + A′ = $O_{n \times n}$, then $|I + \lambda A^2| \geq M$ where 3M + 1 is _____

55. Consider a equation $x^3 - 3x = \sqrt{x+2}$ such that x is a real number then sum of its positive roots is λ where $\left[\frac{\lambda}{2}\right]$ is _____

56. Consider A and B as 2 $\times$ 2 matrices with determinant equal to 1, then $tr(AB) – tr(A)\cdot tr(B) + tr(AB^{-1}) + 2$ is _____

57. Consider an ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and ABCD be a quadrilateral circumscribing the ellipse. Let S be one of its focii, then $\angle ASB + \angle CSD = \frac{3\pi}{\lambda}$ where, λ is _____

58. Consider a parabola $y^2 = 8x$. If PSQ is a focal chord of the parabola whose vertex is A and focus S, V being the middle point of the chord such that $PV^2 = AV^2 + \lambda \cdot AS^2$ where λ is _____

59. If $\alpha \in (0, 1)$ and f: R $\to$R and $\lim_{x\to\infty} f(x) = 0$, $\lim_{x\to\infty} \frac{f(x) - f(\alpha x)}{x} = 0$, then $\lim_{x\to\infty} \frac{f(x)}{x} = \lambda$ where $2\lambda + 7$ is _____

60. Let f: [a, b] $\to$ R be a function, continuous on [a, b] and twice differentiable on (a, b). If, f(a) = f(b) and f′(a) = f′(b), then consider the equation $f''(x) – \lambda(f'(x))^2 = 0$. For any real λ the equation has atleast M roots where 3M + 1 is _____

Space for rough work